

Work with tanh $\rightarrow$ use xavier $\rightarrow$ good results
" " relu $\rightarrow$ " He init $\rightarrow$ " "

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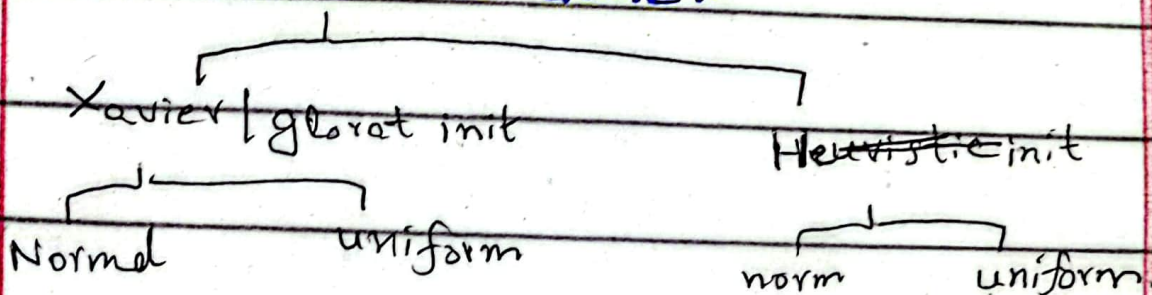
Why weight initialization imp:-

its important as b/c of this $\&$ sigmoid func in 2000 work on deep learning has stopped as it give problems like van. grad, slow convergn etc.

What not to do?

- 1) Zero initialization:- No training.
- 2) Non-zero const value:- Multiple weight at a time update.
- 3) Small random value not initialize.
- 4) Large " " " " " "

What could be done:-



$\rightarrow$ We have randomly initialize weight not with big $\&$ small value.

In xavier when we need a value that is not big or small so in xavier $\frac{1}{\sqrt{n}}$ where n is no of inputs to nodes & this is a solution.

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Small:- `np.random.randn(250, 250) * 0.01`

Large:- `np.random.randn(250, 250) * 1`

Xavier init:-

$$\sqrt{\frac{2}{f_{\text{an}} + f_{\text{out}}}} \quad \# \text{ no of inputs coming towards}$$

$\rightarrow \sum \text{np.random.randn}(2, 2) * \sqrt{\frac{1}{2}}$

He init:- (relu)

$$\sqrt{\frac{2}{f_{\text{an}}}}$$

uniform xavier:- $[-\text{limit}, \text{limit}]$
distribution.

$$\text{limit} = \sqrt{\frac{6}{f_{\text{an}} + f_{\text{out}}}}$$

He uniform:-

$$[-\text{limit}, \text{limit}]$$

$$\text{limit} = \sqrt{\frac{6}{f_{\text{an}}}}$$

By default global uniform is set in keras

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Batch normalization:-

It is a technique using which you can speed up your neural net training.

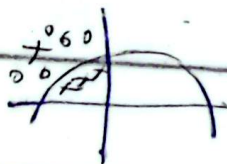
→ Algorithmic method.

→ You can normalized the output of a hidden layer which is input to another layer means makes $\text{mean} = 0$, $\text{stddev} = 1$.

Why use batch norm:-

To minimize cost function & reach local minima.

Covariate shift:- Sometimes distrib-



ution of data changes, input output relation is still same.

→ You still have to retrain ml model.

$\gamma = 1, \beta = 0$
in keras.

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Internal covariate shift:-

Change in distribution of network activations due to change in network parameters during training.

e.g:- meize $\rightarrow$ squeeze $\rightarrow$ sheeize 😊

Intuition b/d batch normalization:-

After every hidden layer output converts to gaussian distribution

$\rightarrow$ Due to this internal covariate shift decreases, $\rightarrow$ γ training fast & stable.

$\rightarrow$ low learning rate.

How it works:-

$\rightarrow$ Apply with mini-Batch GD.

$\rightarrow$ Layer by layer basis apply.